HOPE LAB / METHOD REFERENCE

Microbial Culture Media, Colony Isolation, Purification and Preservation

A supervised method-selection and quality-recording guide for microbiology teaching and research.

HOPE Lab student reference. This document is maintained as a convenient reference for HOPE Lab students and members. It is not an approved institutional SOP and does not replace supervisor instruction, biosafety review, chemical risk assessment, instrument-specific training, or current institutional procedures. Wet-lab work must be conducted only under an approved, supervisor-directed protocol using appropriately assessed materials and facilities.

Purpose and scope

This guide explains how to select and document culture media, understand the distinction between colony isolation and enumeration, assess purity, and plan traceable preservation. It is not an executable culture protocol.

Method principles

- Culture-medium selection must be matched to the identity and known requirements of the approved organism, specimen type, and research question. A single general-purpose medium is not suitable for every organism or purpose.
- Streak plating is a qualitative separation approach intended to obtain spatially separated colonies from a mixed or dense inoculum. Spread plating is commonly used to distribute a measured sample across a surface for viable-count estimation. These approaches answer different questions and should not be treated as interchangeable.
- Purification means obtaining and verifying a culture that is sufficiently uniform for the intended downstream study. Colony appearance alone does not establish species identity or biological safety.
- Preservation strategy depends on organism characteristics, intended storage duration, and the requirements of a recognized culture collection. Cryopreservation and lyophilization are established collection approaches, but their suitability is organism-specific.

Supervised workflow: decision and documentation sequence

- Confirm the organism identity, provenance, intended use, and biosafety classification with the responsible supervisor before any culture work is planned.
- Select a published or culture-collection-recommended medium and method that match the authenticated organism and research objective. Record the source and version of the method.
- Use only the current institutional SOP for preparation, sterilization, inoculation, incubation, colony separation, and any subsequent culture manipulation. This guide intentionally does not supply operational conditions.
- Document the observed colony morphology using a consistent descriptive vocabulary and image record. Treat morphology as a preliminary observation, not definitive identification.
- Confirm culture purity and identity using the laboratory-approved verification method before interpreting downstream results.
- For preservation, follow the organism-specific collection or institutional SOP. Record the stock identifier, provenance, preservation method, date, responsible operator, storage location, and recovery/quality-check status.

Quality controls and records

- Media sterility control: an uninoculated quality-control unit assessed under the approved media-QC procedure.

- Media performance control: an approved reference organism or other validated control appropriate to the medium and its intended claim.
- Identity and purity evidence: retain the method, result, date, and any limitations of the identification process.
- Stock traceability: use unique identifiers and maintain a master/working-stock distinction according to institutional practice. Track passages and any deviations.
- Do not infer purity, safety, or species identity from colony colour, size, edge, elevation, or texture alone.

Interpretation limits

Isolation is not identification. A single colony may still contain more than one organism, and phenotypic appearance can vary with medium, culture age, and environmental conditions. Preservation success must be confirmed by an approved recovery and identity check; it should not be assumed from storage alone.

References and authoritative resources

1. Sanders ER. Aseptic Laboratory Techniques: Plating Methods. Journal of Visualized Experiments. 2012;(63):3064. doi:10.3791/3064. This article distinguishes streak, spread, pour, overlay, and replica plating and emphasizes biosafety classification.

2. Elbing K, Brent R. Growth on solid media. Current Protocols in Molecular Biology. 2002; Unit 1.3. doi:10.1002/0471142727.mb0103s59.

3. Bajerski F, Barheine A, Belke L, Pradella S. Preservation of Microorganisms for Storage in Culture Collections. Methods in Molecular Biology. 2026;3060:505-518. doi:10.1007/978-1-0716-5416-3_21.

4. ATCC. Bacteriology Culture Guide. Culture-collection guidance on media, maintenance, and preservation. https://www.atcc.org/en/resources/culture-guides/bacteriology-culture-guide

Document status: literature-informed educational reference. Confirm current editions, organism-specific applicability, and institutional requirements before use. Prepared 10 November 2026.